

2-2brn:

$$X_{3j} = X_{1j} - X_{14} \quad j = 5, \dots, n-1$$

$$X_{3j} = X_{2j} - X_{24} \quad j = 5, \dots, n-1$$

$$X_{3n} = \quad - X_{1n}$$

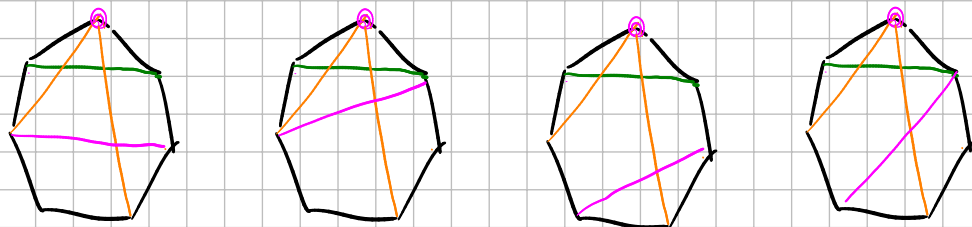
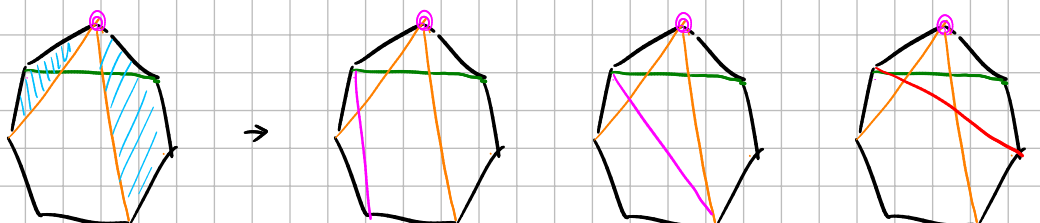
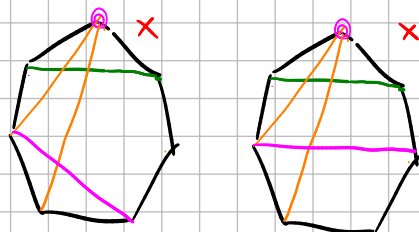
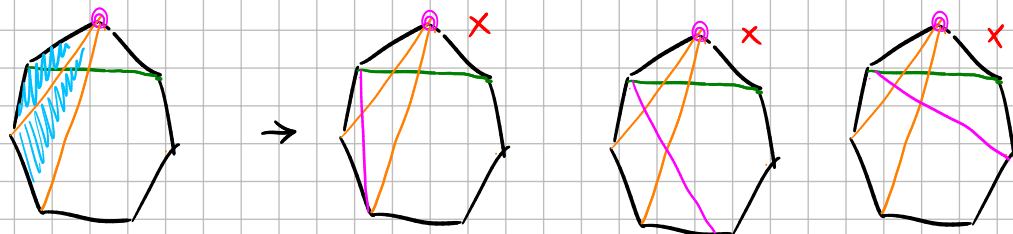
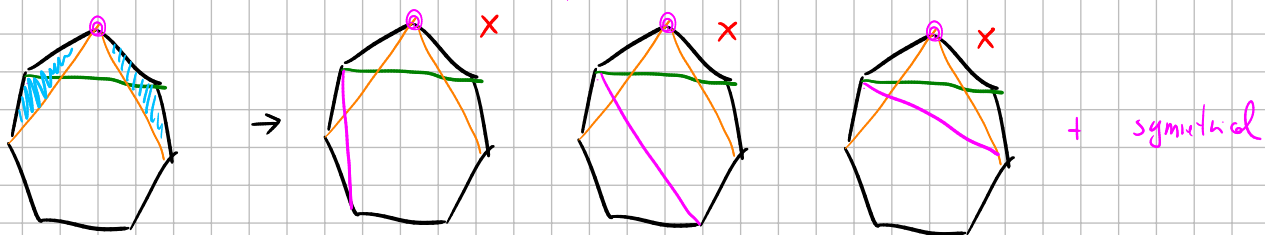
$$X_{3n} = X_{2n} - X_{24}$$

13 is brn free.

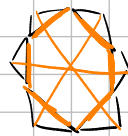
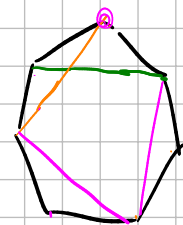
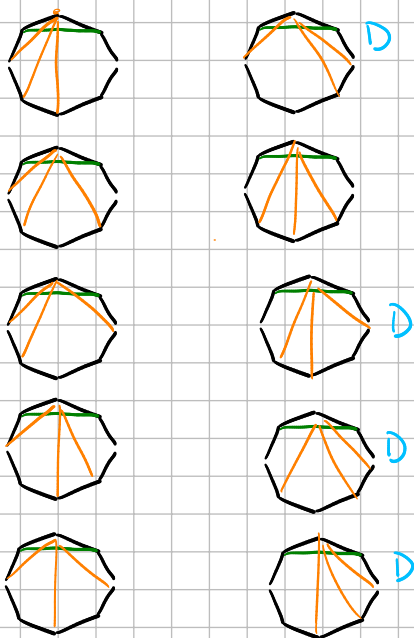
2 can connect to $n-3$ things but we have $n-4$ chords left

• if we use $n-4$ crossing chords then the diagram vanishes (1 step removed)

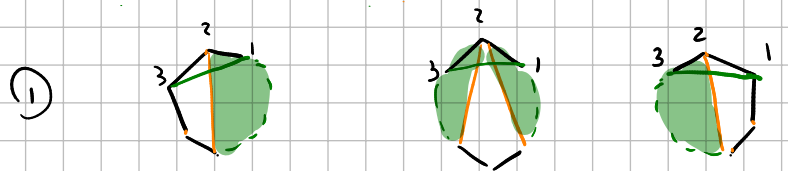
• if we use $n-5$ crossing chords:



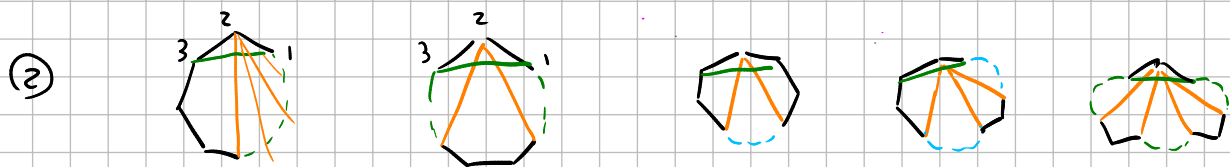
$n=8$
 $C^{(1)}$



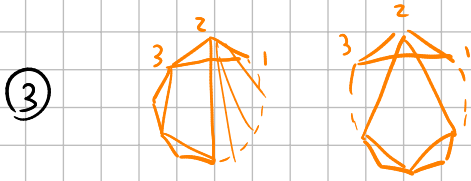
like steps removed, but it's steps removed from a fan



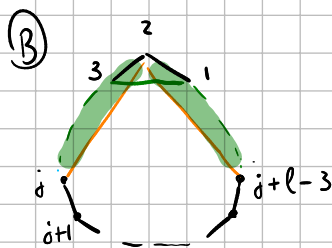
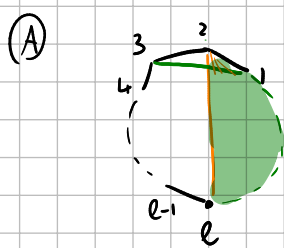
nothing left to add $n-4$ crossings



$n-5$ crossings



2 to add in



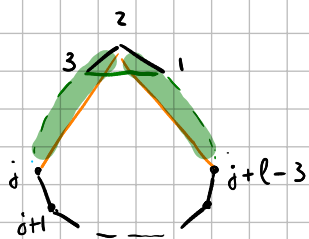
$j \in \{4, \dots, n-1\}$

$l-5$ things to add in
 $(l-1)$ gon

need to prove any way we put $l-5$ chords back in is nilled
 by some $n=0$

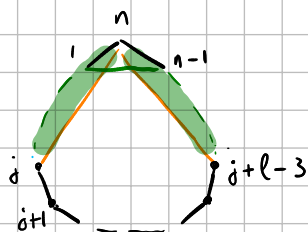
removed
 $n-l+1$ crossings

they must connect to 2 but can cross the orange lines.



① must satisfy 1-zeros

the edges $(i, i+1), (i+1, i+2), \dots$ etc must all be part of a triangle.



$$\frac{1}{X_{i,n-1}} \prod_{i=2}^{\dot{i}} \frac{1}{X_{ni}} \prod_{r=i+l-3}^{n-2} \frac{1}{X_{nr}} \underbrace{C_{\dot{i}}^{(l-5)}(1, \dots, n-1)}_{\downarrow} \quad i \in \{1, n-l+2\}$$

linear extension of diagrams
with $l-5$ chords only involving vertices $(1, \dots, n-1)$

how do we show these vanish?

- find all κ -zero where $\frac{1}{X_{i,n-1}} \prod_{i=2}^{\dot{i}} \frac{1}{X_{ni}} \prod_{r=i+l-3}^{n-2} \frac{1}{X_{nr}}$ can be independent so we can take the residue.

- Using only these κ -zeros show that $C_{\dot{i}}^{(l-5)}(1, \dots, n-1) = 0$

13	14	15	16	17	18
	24	25	26	27	28
		35	36	37	38
			46	47	48
				57	58
					68

24	25	26	27	28	21
	35	36	37	38	31
		46	47	48	41
			57	58	51
				68	61
					71

35	36	37	38	31	32
	46	47	48	41	42
		57	58	51	52
			68	61	62
				71	72
					82

46	47	48	41	42	43
	57	58	51	52	53
		68	61	62	63
			71	72	73
				82	83
					13

57	58	51	52	53	54
	68	61	62	63	64
		71	72	73	74
			82	83	84
				13	14
					24

13	14	15	16	17	•
	24	25	26	27	28
		35	36	37	38
			46	47	48
				57	58
					68

if we pick 2n we can have 35, ..., 3n-1 ides

otherwise pick one from each $(1_j, 2_j, 3_j)$, $j \in \{s, \dots, n-1\}$

$$X_{24} = X_{14} - X_{13}$$

$$x_{25} = -x_{13}$$

$$X_{24} = X_{14} - X_{25}$$

$$x_{13} = -x_{25}$$

3-Zeros

13	14	15	16	17	•
	24	25	26	27	28
		35	36	37	38
			46	47	48
				57	58
					68

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
2	4	6	8	10	12	14	16	15	13	11	9	7	5	3	1
3	6	9	12	15	5	8	11	14	4	7	10	13	16	2	15
4	8	12	16	5	9	13	2	6	10	14	3	7	11	15	4
5	10	15	5	15	10	5	15	10	5	15	10	5	15	10	5
6	12	5	9	13	2	6	10	14	4	7	10	13	16	2	15
7	14	8	11	4	7	10	14	4	7	10	14	4	7	10	14
8	16	1	3	7	11	3	7	11	3	7	11	3	7	11	3
9	15	13	11	4	6	10	14	4	7	10	14	4	7	10	14
10	5	9	13	2	6	10	14	4	7	10	14	4	7	10	14
11	9	11	4	6	10	14	4	7	10	14	4	7	10	14	4
12	13	7	11	3	7	11	3	7	11	3	7	11	3	7	11
13	7	11	3	7	11	3	7	11	3	7	11	3	7	11	3
14	11	3	7	11	3	7	11	3	7	11	3	7	11	3	7
15	4	6	10	14	4	7	10	13	16	2	15	4	6	10	14
16	3	15	14	5	13	2	6	10	14	4	7	10	13	16	2

$$\begin{aligned} & \{(X_{1,6}, X_{4,7}), (X_{1,7}, X_{4,6}), (X_{2,5}, X_{3,8}), (X_{2,6}, X_{3,7}), (X_{2,7}, X_{3,6}), (X_{2,8}, X_{3,5}), (X_{1,6}, X_{2,5}, X_{3,7}), (X_{1,6}, X_{2,7}, X_{3,5}), (X_{1,6}, X_{2,7}, X_{3,7}), \\ & (X_{1,7}, X_{2,5}, X_{3,6}), (X_{1,7}, X_{2,6}, X_{3,5}), (X_{1,7}, X_{2,6}, X_{3,6}), (X_{2,5}, X_{3,6}, X_{3,7}), (X_{2,6}, X_{2,7}, X_{3,5}), (X_{2,6}, X_{2,7}, X_{3,8}), (X_{2,6}, X_{3,6}, X_{4,7}), (X_{2,6}, X_{3,8}, X_{4,7}), \\ & (X_{2,7}, X_{3,7}, X_{4,6}), (X_{2,7}, X_{3,8}, X_{4,6}), (X_{2,8}, X_{3,6}, X_{3,7}), (X_{2,8}, X_{3,6}, X_{4,7}), (X_{2,8}, X_{3,7}, X_{4,6}), (X_{1,6}, X_{1,7}, X_{2,5}, X_{3,5}), (X_{2,8}, X_{3,8}, X_{4,6}, X_{4,7})\} \end{aligned}$$

← everything with 16

$$\{\{X_{2,5}, X_{3,5}, X_{1,6}, X_{1,7}\}, \{X_{3,5}, X_{1,6}, X_{2,7}\}, \{X_{2,5}, X_{1,6}, X_{3,7}\}, \{X_{1,6}, X_{4,7}\}, \{X_{1,6}, X_{2,7}, X_{3,7}\}\}$$

everything with 26

$$\{\{X_{3,5}, X_{1,7}, X_{2,6}\}, \{X_{3,5}, X_{2,6}, X_{2,7}\}, \{X_{2,6}, X_{3,7}\}, \{X_{3,8}, X_{2,6}, X_{2,7}\}, \{X_{3,8}, X_{2,6}, X_{4,7}\}, \{X_{1,7}, X_{2,6}, X_{3,6}\}, \{X_{2,6}, X_{3,6}, X_{4,7}\}\}$$

25

$$\{\{X_{2,5}, X_{3,5}, X_{1,6}, X_{1,7}\}, \{X_{2,5}, X_{1,6}, X_{3,7}\}, \{X_{2,5}, X_{1,7}, X_{3,6}\}, \{X_{2,5}, X_{3,6}, X_{3,7}\}, \{X_{2,5}, X_{3,8}\}\}$$

Gr psych 26, 36, 17

